Raspberry Pi Foundation

Lesson 0: Discovering AI

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This lesson introduces artificial intelligence (AI) as a set of computer tools created by humans to help solve difficult problems. Together with your students, you will explore the core concept behind AI: that modern AI systems work by detecting patterns in data (such as text, images, or numbers) and using those patterns to generate predictions. Students will understand the prediction pathway of 'data–patterns–predictions' and identify various real-world uses of AI systems.

Objectives

Objective 1 Describe how data is used by AI systems to make predictions

Objective 2 Identify problems AI systems can help to solve

Keywords

- AI system
- Data
- Pattern
- Prediction

Lesson outline

Please note that the activities are labelled in the top corner of the slide deck to help you navigate the lesson.

Activity	Objective	Description	Time
Activity 1: Discovering AI	1, 2	Students define AI with a partner, then decide if various modern technologies use AI, indicating their choice by moving to different sides of the room.	15 mins
Activity 2: What are data and predictions?	1	Students explore what is meant by a prediction and work through a scenario, predicting a match outcome based on past score data.	10 mins
Activity 3: The prediction pathway	1, 2	Students explore the three-step process for creating AI systems (data–patterns–predictions). Students are asked to match up the data and prediction to the correct type of AI system.	15 mins
Exit ticket	1, 2	This is an opportunity to assess if the objectives have been met as students decide if systems like voice recognition, a vending machine, and a traffic light use AI and should justify their reasoning for at least one of the examples.	5 mins

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Resources

- Writing equipment

Printed/digitally distributed

- Worksheet 1: The prediction pathway
- Worksheet 2: Exit ticket

Links

- N/A

Subject knowledge

For this lesson, you will need a basic understanding of how data patterns are identified by AI systems and used to make predictions. You should be aware of a variety of example AI systems.

Assessment opportunities

Objective	Description
1	**Activity 3: The prediction pathway**. Students should be able to identify which data is used and what predictions are made by each AI system. There is also an extension activity where students are not given the type of data or prediction to stretch those who finish the task quickly. **Exit ticket**. Students are asked to clearly explain why they think that each system is or is not an AI system.
2	**Exit ticket**. Students are asked to identify which system uses AI. This can be assessed by asking them to write their answers on a worksheet. This can be adapted depending on the needs of your class to assess in a different way, such as by using “think, pair, share” or hands up and down and selecting individuals to justify their answers.

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